

Project 1

Application Deployment

Step 1: Application

Clone the below mentioned repository and deploy the application (Run application in port 3000).

Repo URL : <https://github.com/Vennilavanguvi/Brain-Tasks-App.git>

Created the DevOpsProjects folder and navigated to the command prompt and executed the command as below “git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git”

```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects>git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 1.23 MiB/s, done.

C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects>
```

```
C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App>dir
Volume in drive C is OS
Volume Serial Number is FC02-F590

Directory of C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App

22-07-2026  12:47    <DIR>          .
22-07-2026  12:47    <DIR>          ..
22-07-2026  12:47    <DIR>          dist
22-07-2026  12:47                1,801 README.md
               1 File(s)                1,801 bytes
               3 Dir(s) 260,132,868,096 bytes free
```

Docker:

- Dockerize the application by creating Dockerfile
- Build an application and check output using docker image

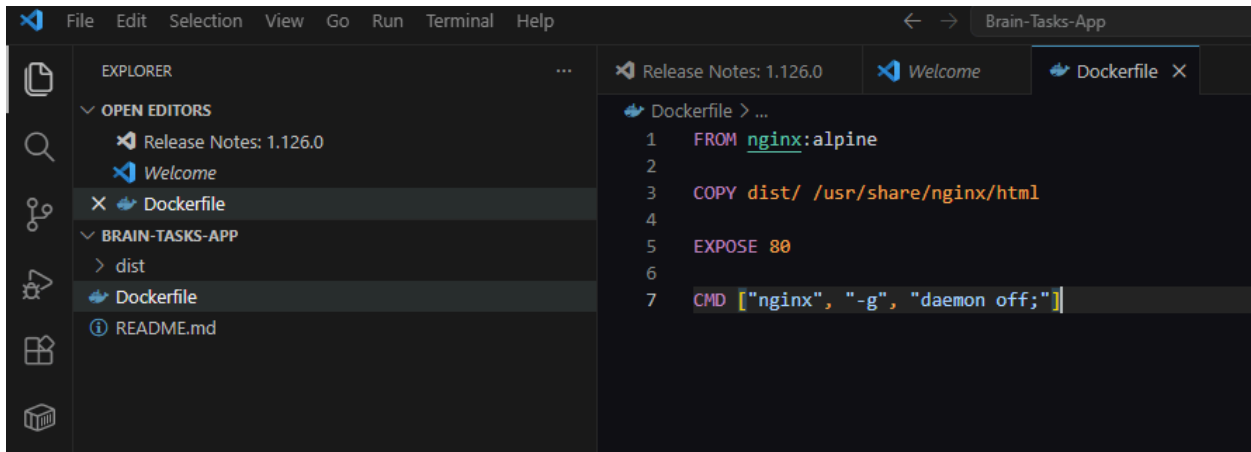
Open VS code. Create a new file named “**Dockerfile**” under BRAIN-TASKS-APP to dockerize the application while executing

FROM nginx:alpine

COPY dist/ /usr/share/nginx/html

EXPOSE 80

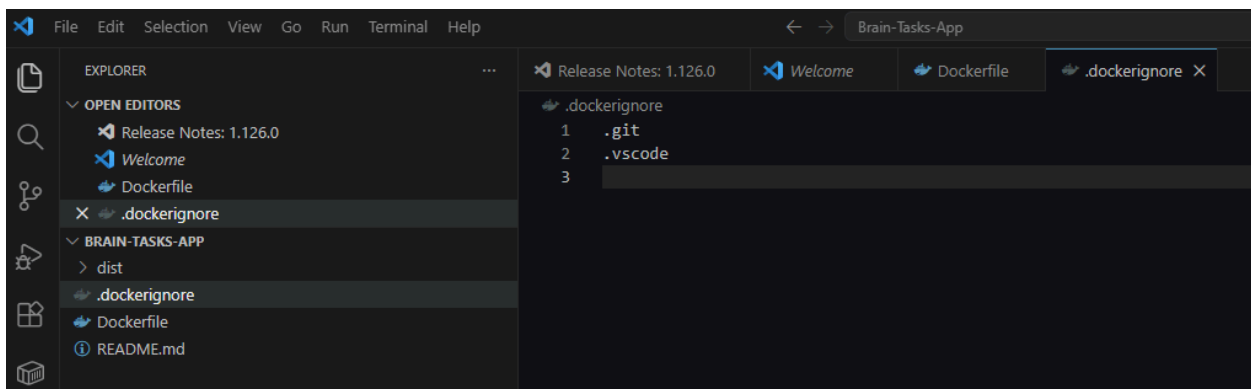
CMD ["nginx", "-g", "daemon off;"]



Save this file and create another file named “**.dockerignore**” and execute the command as below

.git

.vscode



Click on the New terminal and check the docker status by executing `docker ps` and then execute below command

“docker build -t brain-tasks-app:v1 .”

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> docker build -t brain-tasks-app:v1 .
[+] Building 13.0s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 141B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 55B
=> [internal] load build context
=> => transferring context: 318.06kB
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:4a73073bd557c65b759505da037898b61f1be6cbcc3c2c3aeac22d2a470c1752
=> => resolve docker.io/library/nginx:alpine@sha256:4a73073bd557c65b759505da037898b61f1be6cbcc3c2c3aeac22d2a470c1752
=> => sha256:46519e7231d2eb5604df229beb44d59719a489eaa7aca52982535a010b07a9ed 20.31MB / 20.31MB
=> => sha256:390dc935348d8070e695fbaae2a4bb114fb9e69c59f628e7576036ee9d5244c9 1.40kB / 1.40kB
=> => sha256:d0008c891db48b5f526d914bce9e8d889fe1a9d1f08291ae03fe97f871726f38 1.21kB / 1.21kB
=> => sha256:46f977ee452f4399c208714afa034868d6056864f8a0cf3c643ab143dd802c80 404B / 404B
=> => sha256:62bec68d7c31c4c8a19d812d84da5f7748e54690c037979945b6c5b6c924b142 957B / 957B
=> => sha256:1223f016b4e4a2c21f7c49d4837fbfd47a9da6436b511690ca1e582fc2810d59 627B / 627B
=> => sha256:55afa1ecc21d2bb5e5045f32dafee56272ffdb89860bac26f6c32123439af26a4 3.85MB / 3.85MB
=> => sha256:3cd534fe98c64d68a1f4f1c83abb8d5cba7ecfd7be88e592389929d12e6253da 1.89MB / 1.89MB
=> => extracting sha256:55afa1ecc21d2bb5e5045f32dafee56272ffdb89860bac26f6c32123439af26a4
```

To check the image is built, we need to execute “docker images”

```
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> docker images

IMAGE                ID                DISK USAGE    CONTENT SIZE    EXTRA
brain-tasks-app:v1   732f88baa472      93.1MB        26.2MB
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> |
```

Next step is to run the container using below command “**docker run -d -p 3000:80 --name brain-tasks-container brain-tasks-app:v1**”

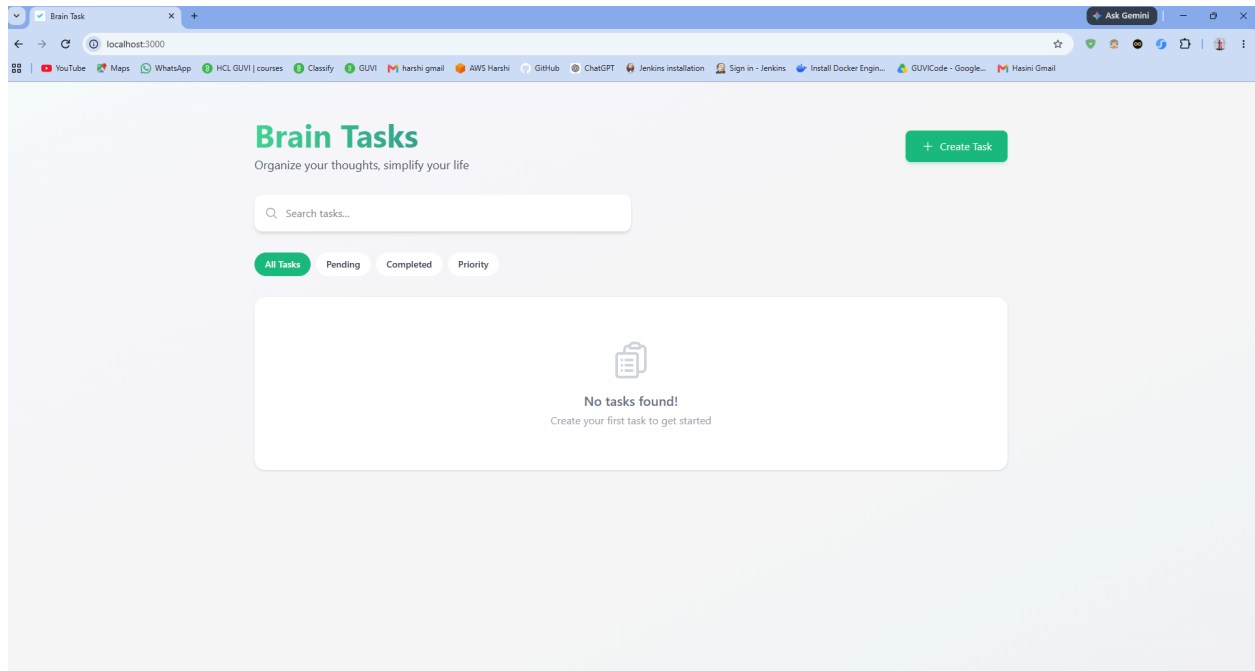
```
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> docker run -d -p 3000:80 --name brain-tasks-container brain-tasks-app:v1
f28c7a3161b99cb08a3223aeb82cb3392856f1c03b7555715f92aa9b116fc514
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> |
```

To verify the container, we need to execute the command “docker ps”

```
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> docker ps

CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS        PORTS                               NAMES
f28c7a3161b9   brain-tasks-app:v1 "/docker-entrypoint..." 2 minutes ago  Up 2 minutes  0.0.0.0:3000->80/tcp, [::]:3000->80/tcp  brain-tasks-container
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> |
```

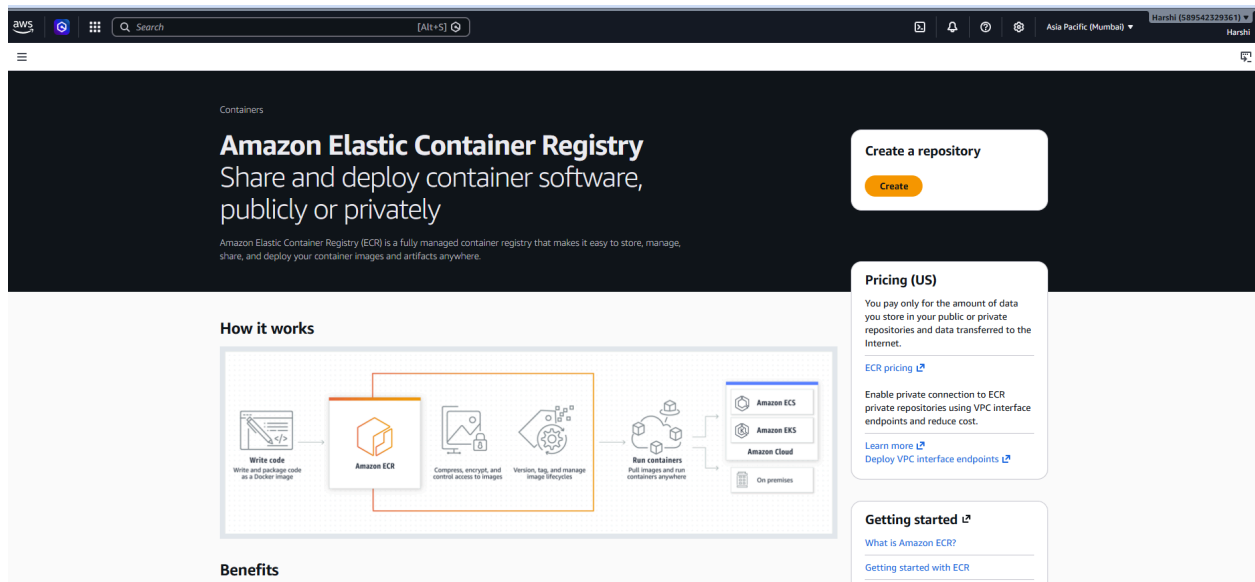
Now we need to launch the application in the browser using “<http://localhost:3000>”



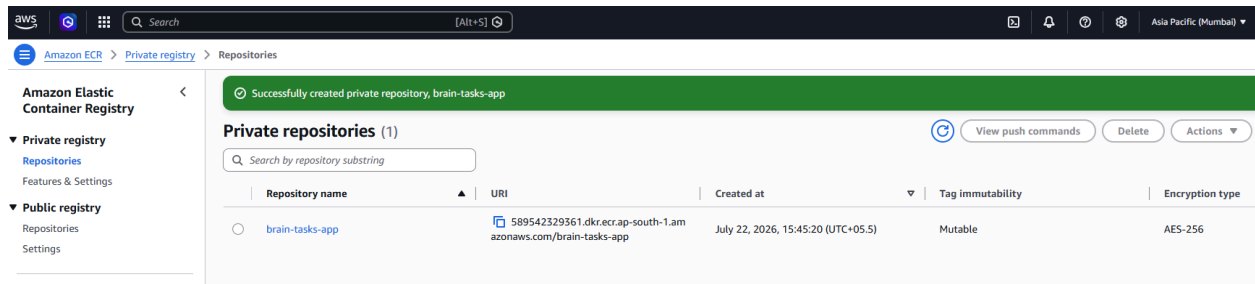
Registry:

- Create an AWS ECR or Dockerhub repository for storing docker images.

Login to AWS console, select the region as Asia Pacific (ap-south-1) and search “ECR” in the Search menu and click on the “**Create a repository**” option in the life panel.



Give the repository name as “**brain-tasks-app**” and click on “**Create**” option



Create the access key from the AWS Console and execute in the Powershell as shown below

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\harin> aws configure
AWS Access Key ID [*****dPqG]: AKIAYSQ3WBQIXB5FB6G6
AWS Secret Access Key [*****dPqG]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\harin> aws sts get-caller-identity
{
  "UserId": "589542329361",
  "Account": "589542329361",
  "Arn": "arn:aws:iam:589542329361:root"
}

PS C:\Users\harin>
```

Now we need to login docker to ECR using this command “**aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 589542329361.dkr.ecr.ap-south-1.amazonaws.com**”

```
PS C:\Users\harin> aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 589542329361.dkr.ecr.ap-south-1.amazonaws.com
Login Succeeded
PS C:\Users\harin>
```

Next step is to tag the docker image using below command “**docker tag brain-tasks-app:v1 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest**” and verifying through “**docker images**” command

```
PS C:\Users\harin> docker tag brain-tasks-app:v1 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
PS C:\Users\harin> docker images
```

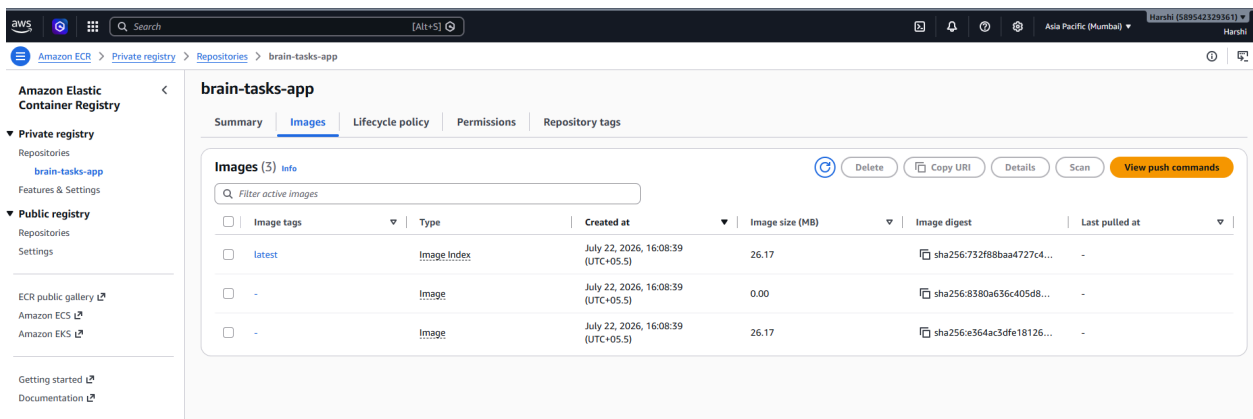
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest	732f88baa472	93.1MB	26.2MB	U
brain-tasks-app:v1	732f88baa472	93.1MB	26.2MB	U

```
PS C:\Users\harin> |
```

Now we need to push the image using “**docker push** 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest” command

```
PS C:\Users\harin> docker push 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app]
1223f016b4e4: Pushed
3cd534fe98c6: Pushed
46f977ee452f: Pushed
46519e7231d2: Pushed
2644ca801f6b: Pushed
55afa1ecc21d: Pushed
d0008c891db4: Pushed
390dc935348d: Pushed
62bec68d7c31: Pushed
ff1cfbf5ceb6: Pushed
latest: digest: sha256:732f88baa4727c41f191a65854c1177cc9edab00ade2e1eb503f13e0cca82f1e size: 856
PS C:\Users\harin> |
```

Now verify that the image is pushed to ECR in the AWS console



Kubernetes:

- Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.
- Write deployment and service YAML files.
- Deploy using kubectl via CodeBuild or Codepipeline.

First we need to install kubectl using the command “**curl.exe -LO** <https://dl.k8s.io/release/v1.31.0/bin/windows/amd64/kubectl.exe>”

```
PS C:\Users\harin> curl.exe -LO "https://dl.k8s.io/release/v1.31.0/bin/windows/amd64/kubectl.exe"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
             Dload  Upload   Total     Spent    Left     Speed
100 55.20M 100 55.20M    0      0  3.46M      0  00:15   00:15   00:15   2.18M
PS C:\Users\harin> mkdir c:\kubectl

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          22-07-2026         16:15         kubectl

PS C:\Users\harin> move kubectl.exe c:\kubectl
PS C:\Users\harin> setx PATH "$($env:PATH);C:\kubectl"

SUCCESS: Specified value was saved.
PS C:\Users\harin> |
```

```
PS C:\Users\harin> kubectl version --client
Client Version: v1.33.2
Kustomize Version: v5.6.0
PS C:\Users\harin>
```

Now we need to install eksctl using the command “**winget install Weaveworks.eksctl**”

```
PS C:\Users\harin> eksctl version
0.229.0
PS C:\Users\harin>
```

Now we need to create EKS Cluster using this command “**eksctl create cluster --name brain-cluster --region ap-south-1 --nodegroup-name workers --node-type t2.micro --nodes 2**”

```
Windows PowerShell
eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" add-on is via pod identity associations; after add-on creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update add-on'
2026-07-22 20:06:50 [i] creating add-on: vpc-cni
2026-07-22 20:06:50 [i] successfully created add-on: vpc-cni
2026-07-22 20:06:51 [i] creating add-on: kube-proxy
2026-07-22 20:06:51 [i] successfully created add-on: kube-proxy
2026-07-22 20:06:51 [i] creating add-on: coredns
2026-07-22 20:06:52 [i] successfully created add-on: coredns
2026-07-22 20:08:53 [i] building managed nodegroup stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:08:53 [i] deploying stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:08:53 [i] waiting for CloudFormation stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:09:23 [i] waiting for CloudFormation stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:10:19 [i] waiting for CloudFormation stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:11:47 [i] waiting for CloudFormation stack "eksctl-brain-cluster-nodegroup-workers"
2026-07-22 20:11:47 [i] waiting for the control plane to become ready
2026-07-22 20:11:51 [✓] saved kubeconfig as "C:\\Users\\harin\\.kube\\config"
2026-07-22 20:11:51 [i] no tasks
2026-07-22 20:11:51 [✓] all EKS cluster resources for "brain-cluster" have been created
2026-07-22 20:11:51 [i] nodegroup "workers" has 1 node(s)
2026-07-22 20:11:51 [i] node "ip-192-168-29-146.ap-south-1.compute.internal" is ready
2026-07-22 20:11:51 [i] waiting for at least 1 node(s) to become ready in "workers"
2026-07-22 20:11:51 [i] nodegroup "workers" has 1 node(s)
2026-07-22 20:11:51 [i] node "ip-192-168-29-146.ap-south-1.compute.internal" is ready
2026-07-22 20:11:51 [✓] created 1 managed nodegroup(s) in cluster "brain-cluster"
2026-07-22 20:11:51 [i] creating add-on: metrics-server
2026-07-22 20:11:52 [i] successfully created add-on: metrics-server
2026-07-22 20:11:54 [i] kubectl command should work with "C:\\Users\\harin\\.kube\\config", try 'kubectl get nodes'
2026-07-22 20:11:54 [✓] EKS cluster "brain-cluster" in "ap-south-1" region is ready
PS C:\\Users\\harin>
```

To verify the cluster, we use this command “**kubectl get nodes**”

```
PS C:\\Users\\harin> kubectl get nodes
NAME
ip-192-168-29-146.ap-south-1.compute.internal   Ready   <none>   7m3s   v1.34.9-eks-8f14419
PS C:\\Users\\harin> kubectl cluster-info
Kubernetes control plane is running at https://48639183BC8927CE261F12FC93A74931.gr7.ap-south-1.eks.amazonaws.com
CoreDNS is running at https://48639183BC8927CE261F12FC93A74931.gr7.ap-south-1.eks.amazonaws.com/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
PS C:\\Users\\harin> kubectl get pods -A
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE
kube-system   aws-node-74jz4                         2/2     Running   0           7m47s
kube-system   coredns-8477c579f6-bg9bm              1/1     Running   0           11m
kube-system   coredns-8477c579f6-z5zm9              1/1     Running   0           11m
kube-system   kube-proxy-h95g7                      1/1     Running   0           7m47s
kube-system   metrics-server-8b59c5fc6-2csjv        0/1     Pending   0           6m8s
kube-system   metrics-server-8b59c5fc6-vn4zc        0/1     Pending   0           6m8s
PS C:\\Users\\harin>
```

To deploy the Nginx, we use the below command “**kubectl create deployment nginx --image=nginx**”

```

PS C:\Users\harin> kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
PS C:\Users\harin> kubectl get deployments
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
nginx     0/1     1            0           58s
PS C:\Users\harin> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-66686b6766-frbsh             0/1     Pending   0          88s
PS C:\Users\harin> kubectl expose deployment nginx --type=LoadBalancer --port=80
service/nginx exposed
PS C:\Users\harin>

```

```

PS C:\Users\harin> kubectl get nodes
NAME                                STATUS   ROLES    AGE   VERSION
ip-192-168-41-250.ap-south-1.compute.internal Ready    <none>  3m5s  v1.34.9-eks-8f14419
PS C:\Users\harin> kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
PS C:\Users\harin> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-66686b6766-f8rkh             1/1     Running   0          81s
PS C:\Users\harin> kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes ClusterIP  10.100.0.1    <none>        443/TCP     10m
PS C:\Users\harin> kubectl get endpoints nginx
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
Error from server (NotFound): endpoints "nginx" not found
PS C:\Users\harin> kubectl expose deployment nginx --type=LoadBalancer --port=80
service/nginx exposed
PS C:\Users\harin> kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes ClusterIP  10.100.0.1    <none>        443/TCP     11m
nginx     LoadBalancer 10.100.118.198 a53bff01aaa4b468a9e104207c4a9bf6-1873032059.ap-south-1.elb.amazonaws.com 80:30439/TCP 10s
PS C:\Users\harin> kubectl get endpoints nginx
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS          AGE
nginx     192.168.56.236:80 20s
PS C:\Users\harin> |

```

Push the existing docker image using this command “**docker push**
589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest”

```

PS C:\Users\harin> aws configure
AWS Access Key ID [*****B6G6]: AKIAYSQ3WBQIVDSMVAQO
AWS Secret Access Key [*****6qWc]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\harin> aws sts get-caller-identity
{
  "UserId": "589542329361",
  "Account": "589542329361",
  "Arn": "arn:aws:iam::589542329361:root"
}
PS C:\Users\harin> aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 589542329361.dkr.ecr.ap-south-1.amazonaws.com
Login Succeeded
PS C:\Users\harin> docker push 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app]
d0008c891db4: Pushed
390dc935348d: Pushed
55afalecc21d: Pushed
26444ca801feb: Pushed
62bec68d7c31: Pushed
46f077ee452f: Pushed
46519c7231d2: Pushed
ff1cfbf5ceb6: Pushed
3cd534fe98c6: Pushed
1223f016b4e4: Pushed
latest: digest: sha256:732f88baa4727c41f191a65854c1177cc9edab0ade2e1eb503f13e0cca82f1e size: 856
PS C:\Users\harin>

```

Now create the EKS cluster, initially the status will be “**Creating**”, once the status changes to “**Active**”

Amazon Elastic Kubernetes Service > Clusters > brain-tasks-cluster

Amazon Elastic Kubernetes Service

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New: You can now opt in to OTel Container Insights with OpenTelemetry and PromQL support starting with version 6.2.0 of the CloudWatch Observability add-on. [Learn more](#)

Notifications 0 0 0 0 4 0

brain-tasks-cluster

The Amazon EKS console does not show Kubernetes resources until the cluster has an ACTIVE or UPDATING state. Please try again in a few minutes.

▼ Cluster info

Status Creating	Kubernetes version 1.36	Support period Standard support until August 2, 2027	Provider EKS
Cluster health 0	Upgrade insights 0	Node health issues 0	Capability issues 0

Overview Resources Compute Networking Add-ons Capabilities Access Observability Update history & backups Tags

Details

API server endpoint -	OpenID Connect provider URL -	Created a few seconds ago
Certificate authority -	Cluster IAM role ARN arn:aws:iam::589542329361:role/AmazonEKSAutoClusterRole View in IAM	Cluster ARN arn:aws:eks:ap-south-1:589542329361:cluster/brain-tasks-cluster
		Platform version eks.8

Amazon Elastic Kubernetes Service > Clusters > brain-tasks-cluster

Amazon Elastic Kubernetes Service

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New: You can now opt in to OTel Container Insights with OpenTelemetry and PromQL support starting with version 6.2.0 of the CloudWatch Observability add-on. [Learn more](#)

Notifications 0 0 0 0 2 0

brain-tasks-cluster

▼ Cluster info

Status Active	Kubernetes version 1.36	Support period Standard support until August 2, 2027	Provider EKS
Cluster health 0	Upgrade insights 0	Node health issues 0	Capability issues 0

Overview Resources Compute Networking Add-ons Capabilities Access Observability Update history & backups Tags

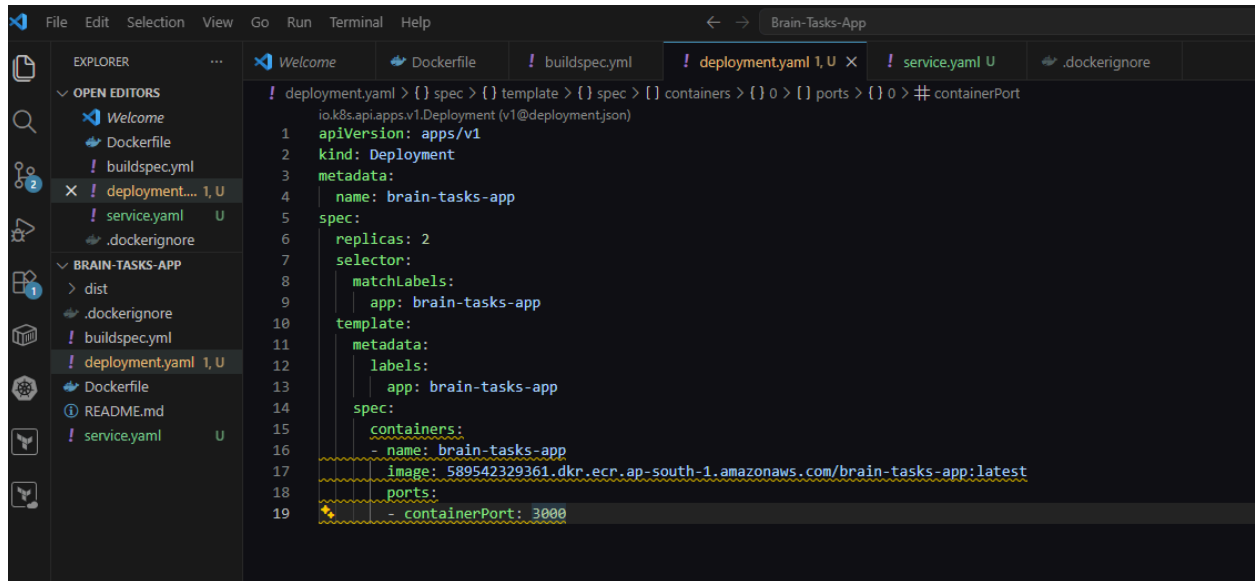
Details

API server endpoint https://DE0A75F110CC148BD97BE69824A52F6C.gr7.ap-south-1.eks.amazonaws.com	OpenID Connect provider URL https://oidc.eks.ap-south-1.amazonaws.com/id/DE0A75F110CC148BD97BE69824A52F6C	Created 14 minutes ago
Certificate authority LS0tLS1CRUdJTiBDRVJUSUZJQ0FURSB0LS0tLS1SUCVENDQWUyZDZ3SjUJZ0U5kh15XFLVXcwT1V3RFFZSkvWklodmNOQVFFTEJRQXGdGVVUTUJFR0EXVUUKQXhNS2EzVmluWEpTWlhbSGN6Q	Cluster IAM role ARN arn:aws:iam::589542329361:role/AmazonEKSAutoClusterRole View in IAM	Cluster ARN arn:aws:eks:ap-south-1:589542329361:cluster/brain-tasks-cluster
		Platform version eks.8

EKS Auto Mode
EKS automates routine cluster tasks for compute, storage, and networking to meet application compute needs.

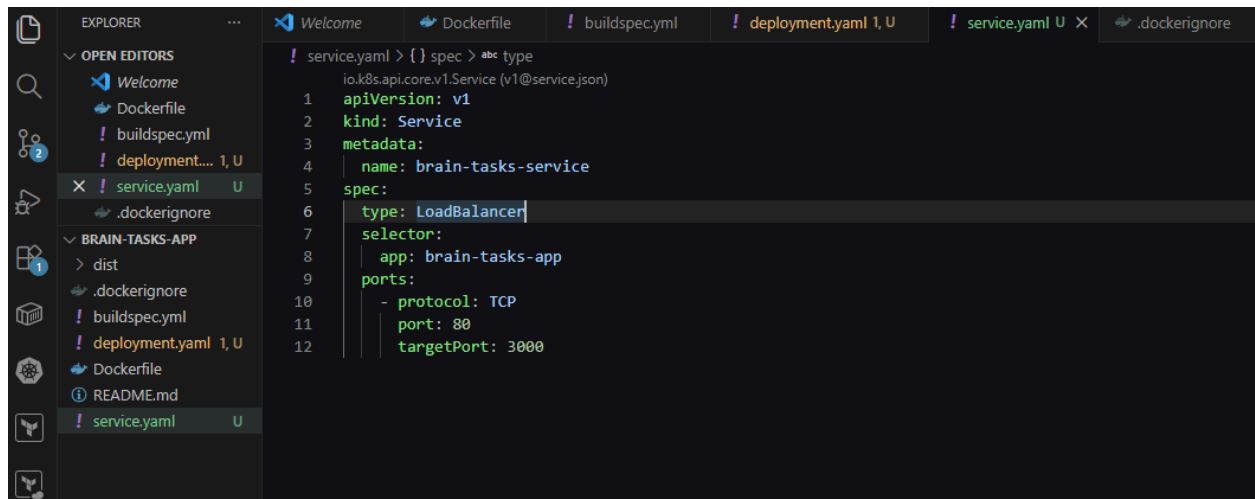
Manage

Creating deployment.yaml and service.yaml files



The screenshot shows the Visual Studio Code interface with the 'deployment.yaml' file open. The Explorer sidebar on the left shows the project structure for 'BRAIN-TASKS-APP', including files like 'dist', '.dockerignore', 'buildspec.yml', 'deployment.yaml', 'Dockerfile', 'README.md', and 'service.yaml'. The main editor displays the 'deployment.yaml' file with the following content:

```
! deployment.yaml > {} spec > {} template > {} spec > {} containers > {} 0 > {} ports > {} 0 > {} containerPort
io.k8s.api.apps.v1.Deployment (v1@deployment.json)
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: brain-tasks-app
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: brain-tasks-app
10   template:
11     metadata:
12       labels:
13         app: brain-tasks-app
14     spec:
15       containers:
16         - name: brain-tasks-app
17           image: 589542329361.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
18           ports:
19             - containerPort: 3000
```



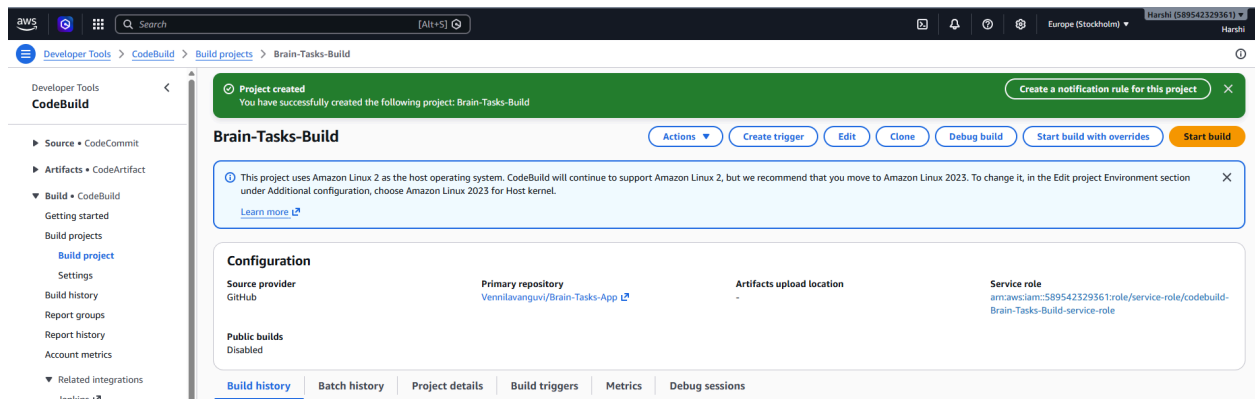
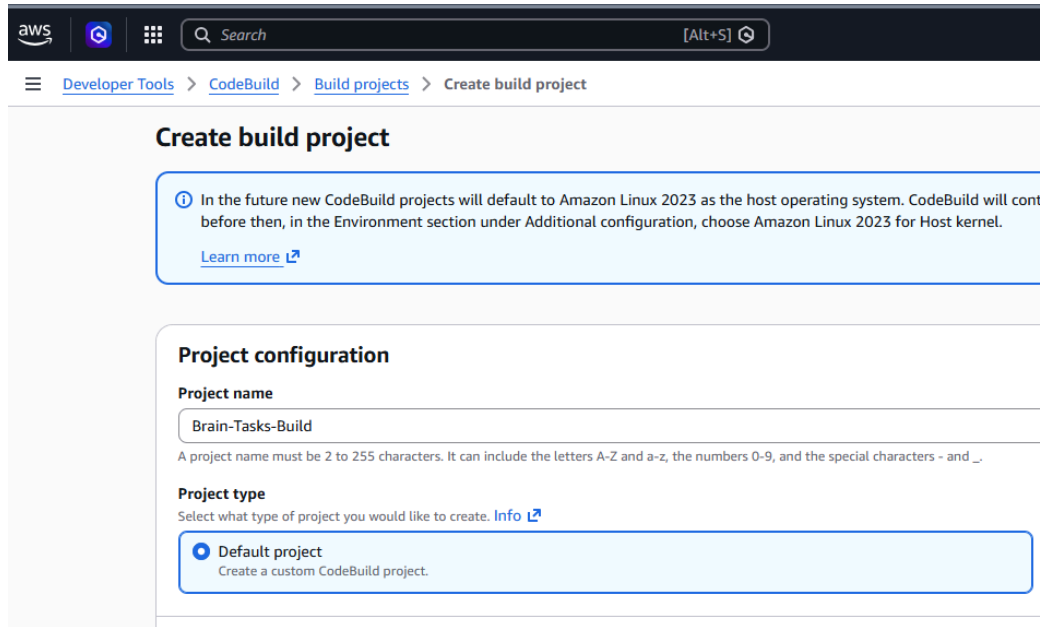
The screenshot shows the Visual Studio Code interface with the 'service.yaml' file open. The Explorer sidebar on the left shows the project structure for 'BRAIN-TASKS-APP', including files like 'dist', '.dockerignore', 'buildspec.yml', 'deployment.yaml', 'Dockerfile', 'README.md', and 'service.yaml'. The main editor displays the 'service.yaml' file with the following content:

```
! service.yaml > {} spec > abc type
io.k8s.api.core.v1.Service (v1@service.json)
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: brain-tasks-service
5  spec:
6    type: LoadBalancer
7  selector:
8    app: brain-tasks-app
9  ports:
10   - protocol: TCP
11     port: 80
12     targetPort: 3000
```

CodeBuild:

- Create a CodeBuild project:
- Source: Connect to your repository
- Environment: Use managed image (Amazon Linux, Ubuntu)
- Write and define commands in buildspec.yml.

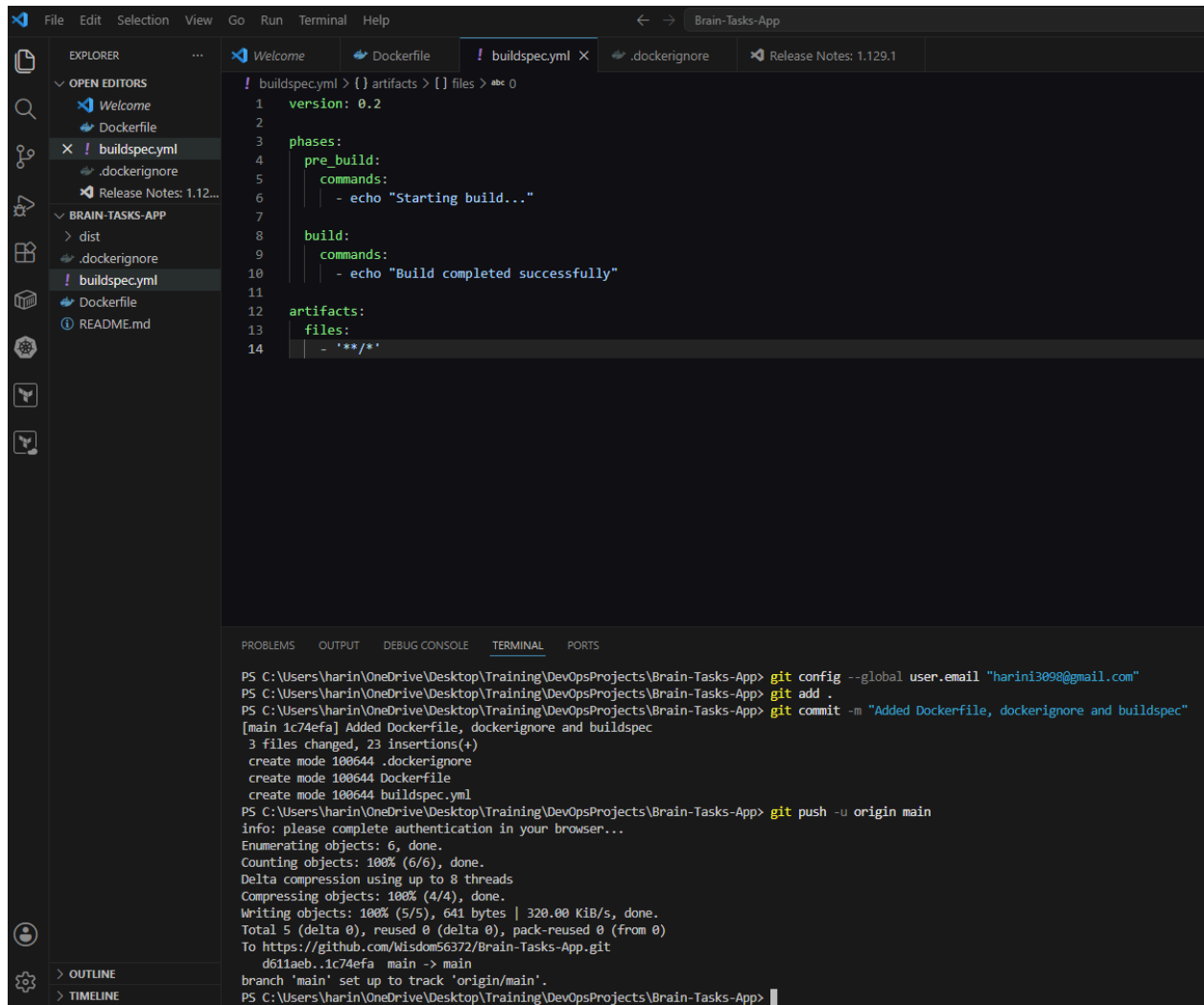
Go to the AWS Console, search for CodeBuild and click on “**Create Project**” option and enter the project name



Version Control:

- Push the codebase to a Git provider (GitHub).
- Use CLI commands to push code.

We need to create a new file called `buildspec.yml` and using the `git add`, `git commit` and `git push` we need to change the Git remote to the repository.

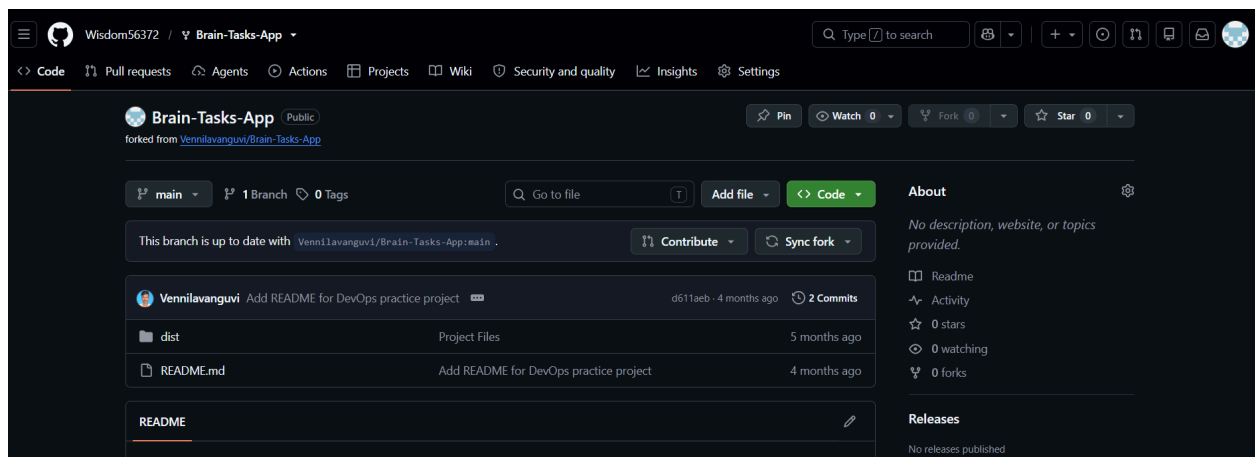


The image shows a Visual Studio Code editor window with the file explorer on the left and the terminal at the bottom. The file explorer shows the project structure for 'Brain-Tasks-App', including files like 'dist', 'buildspec.yml', 'Dockerfile', and 'README.md'. The 'buildspec.yml' file is open in the editor, showing a YAML configuration for a build process. The terminal at the bottom shows the execution of several git commands, including 'git config', 'git add', 'git commit', and 'git push', which successfully push the code to a GitHub repository.

```
! buildspec.yml > {} artifacts > {} files > 0
1  version: 0.2
2
3  phases:
4    pre_build:
5      commands:
6        - echo "Starting build..."
7
8    build:
9      commands:
10       - echo "Build completed successfully"
11
12  artifacts:
13    files:
14      - '**/*'
```

```
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> git config --global user.email "harini3098@gmail.com"
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> git add .
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> git commit -m "Added Dockerfile, dockerignore and buildspec"
[main 1c74efa] Added Dockerfile, dockerignore and buildspec
3 files changed, 23 insertions(+)
create mode 100644 .dockerignore
create mode 100644 Dockerfile
create mode 100644 buildspec.yml
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 641 bytes | 320.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Wisdom56372/Brain-Tasks-App.git
d611aeb..1c74efa  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\harin\OneDrive\Desktop\Training\DevOpsProjects\Brain-Tasks-App>
```

We need to fork the repository <https://github.com/Vennilavanguvi/Brain-Tasks-App.git>



Next step is to start the build by clicking on “**Start build**” in the top-right corner
Build status : “**Succeeded**”

The screenshot shows the AWS CodeBuild console. A green banner at the top indicates "Build started" and "You have successfully started the following build: Brain-Tasks-Build:da851799-8361-440f-a07d-f9ac61985f27". Below this, the build ID "Brain-Tasks-Build:da851799-8361-440f-a07d-f9ac61985f27" is displayed with buttons for "Stop build", "Debug build", and "Retry build".

The "Build status" section shows:

- Status: Succeeded
- Initiator: root
- Build ARN: arn:aws:codebuild:eu-north-1:589542329361:build/Brain-Tasks-Build:da851799-8361-440f-a07d-f9ac61985f27
- Resolved source version: 1c74efa877364b8f50b87d39630edb447a778c8
- Start time: Jul 23, 2026 5:00 PM (UTC+5:30)
- End time: Jul 23, 2026 5:00 PM (UTC+5:30)
- Build number: 1

Below the status, there are tabs for "Build logs", "Phase details", "Reports", "Environment variables", "Build details", and "Resource utilization". The "Build logs" tab is active, showing "Showing the last 47 lines of the build log. View entire log" and a "Tail logs" button. The log content shows the build process starting on CodeBuild On-demand, waiting for agent ping, downloading source, and running the build.

CodePipeline:

- Source: GitHub
- Build: AWS CodeBuild project
- Deploy: Deploy to EKS via Codepipeline EKS deploy stage

Navigate to the AWS console and search for CodePipeline and click on the “**Create pipeline**” option.

The screenshot shows the AWS CodePipeline console. The "Pipelines" page is displayed, showing a search bar and buttons for "View history", "Release change", "Delete pipeline", and "Create pipeline". Below the search bar, there is a table with columns: "Name", "Latest execution status", "Latest source revisions", and "Latest execution started". The table is empty, and a message states "No results. There are no results to display."

Step 3

Configure template

Source provider

This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via GitHub App) ▼

Connection

Choose an existing connection that you have already configured, or create a new one and then return to this task.

arn:aws:codeconnections:eu-  or [Connect to GitHub](#)

Repository name

Choose a repository in your GitHub account.

Wisdom56372/Brain-Tasks-App ✕

You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.

Default branch

Default branch will be used only when pipeline execution starts from a different source or manually started.

main ✕

Output artifact format

Choose the output artifact format.

- ☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.
- ☐ **Full clone**
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions. [Learn more](#)

Cancel

Previous

Next

Trigger configuration

You can add additional pipeline triggers after the pipeline is created.

Trigger type

No filter

Step 4: Add build stage

Build action provider

Build action provider

AWS CodeBuild

ProjectName

Brain-Tasks-Build

Commands

-

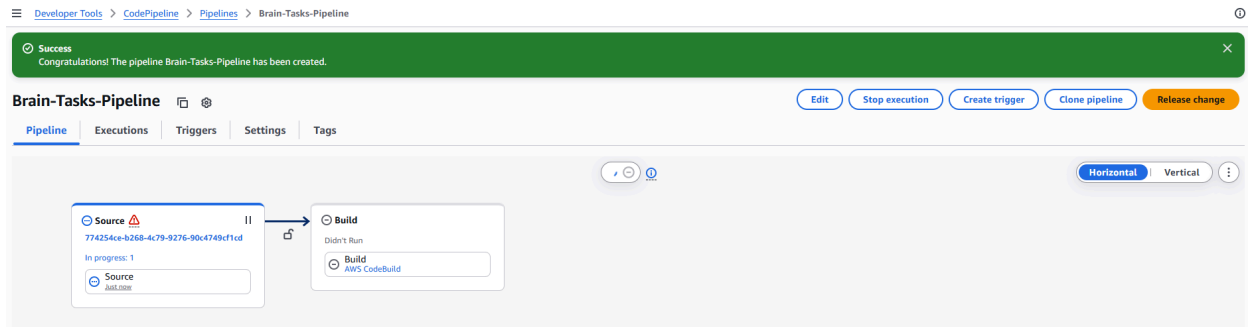
Enable automatic retry on stage failure

Enabled

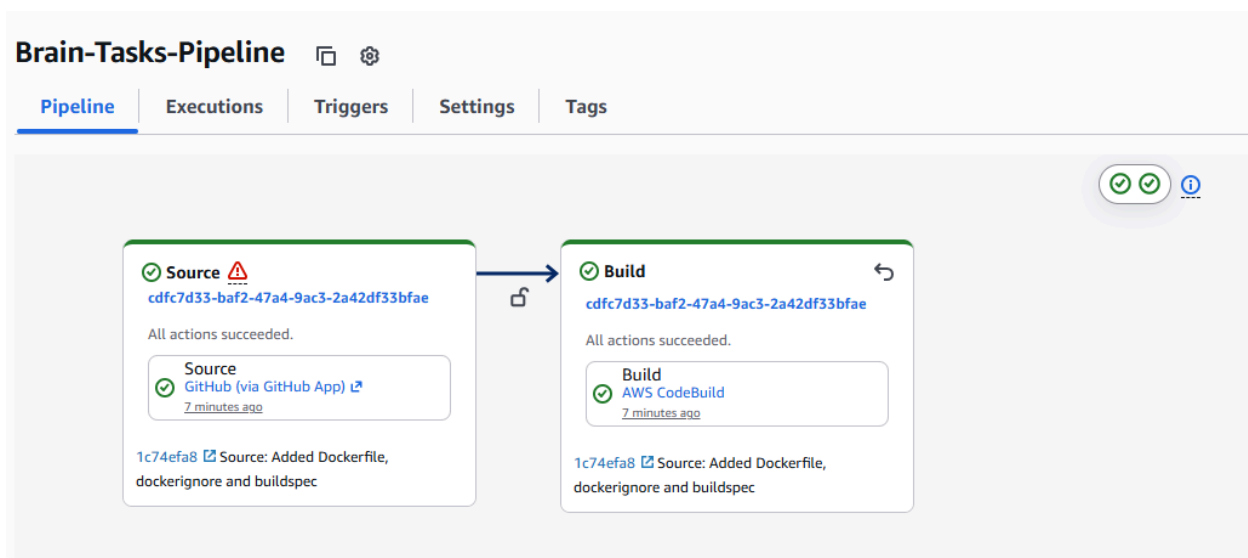
Cancel

Previous

Create pipeline



Next we need to click on “**Release change**”



Monitoring:

- Use CloudWatch Logs to track build, deploy, and application logs.

Search “CloudWatch” in the AWS console and click on the “Logs” icon and select “Log Management” option.

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Analytics

Log Anomalies

Metrics

Network Monitoring

Setup

Observability Workshops

Log groups

Data sources - new

Summary

Get-set-go with unified data store

Log groups (1)

By default, we only load up to 10,000 log groups.

Filter log groups or try pattern search

Exact match

Log group

Log class

Anomaly d...

Deletion pro...

Bearer token...

Data prote...

Sensitive ...

Retention

Metric filters

Co

/aws/codebuild/Brain-Tasks-Build

Standard

Configure

Off

Off

-

-

Never expire

-

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log Analytics

Log Anomalies

Metrics

Network Monitoring

Setup

Observability Workshops

Log management

/aws/codebuild/Brain-Tasks-Build

Log group details

Log class

Info

Standard

ARN

am:awslogs:ap-south-1:589542329361:log-group:/aws/codebuild/Brain-Tasks-Build:*

Creation time

2 hours ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Deletion protection

Off

Data protection

-

Sensitive data count

-

Custom field indexes

Configure

Transformer

Configure

Anomaly detection

Configure

Bearer token authentication

Off

Syslog ingestion

Info

Configure

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field indexes

Transformer

Log streams (2)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match

Show expired

Info

Log stream

Last event time

ce8d92ab-27e0-425b-b0f9-5e3683cc2710

2026-07-23 15:23:37 (UTC)

5bd126b6-20fc-4891-a074-741bc14dd1a8

2026-07-23 14:45:10 (UTC)

CloudWatch <

Favorites and recents ▶

Ingestion

Dashboards

Alarms   

▶ AI Operations

▶ GenAI Observability

▶ Application Signals (APM)

▶ Infrastructure Monitoring

▼ Logs

Log Management

Log Analytics [New](#)

Log Anomalies

▶ Metrics

▶ Network Monitoring

▶ Setup

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q Filter events - press enter to search

Clear1m30m1h12hCustomUTC timezoneDisplay

Timestamp	Message
No older events at this moment. Retry	
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:26.223404 Running on CodeBuild On-demand
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:26.223412 Waiting for agent ping
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:29.804887 Waiting for DOWNLOAD_SOURCE
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:32.735926 Phase is DOWNLOAD_SOURCE
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:32.735969 CODEBUILD_SRC_DIR=/codebuild/output/src3645708076/src
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:32.736411 YAML location is /codebuild/output/src3645708076/src/buildspec.yml
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:32.738181 Setting HTTP client timeout to higher timeout for S3 source
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:32.738518 Processing environment variables
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:34.305047 No runtime version selected in buildspec.
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:34.328275 Moving to directory /codebuild/output/src3645708076/src
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:34.328299 Cache is not defined in the buildspec
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:35.074774 Skip cache due to: no paths specified to be cached
2026-07-23T15:23:37.463Z	[Container] 2026/07/23 15:23:35.074998 Registering with agent